

Compensated Choice and Recursive Counterfactual Identification

Abstract

This paper studies what compensated choice can identify about counterfactual responses in a recursive environment. A primitive binary experiment varies an actual monetary transfer between two current actions at a fixed inherited state. The same monetary operation is used when the primitive pair induces an implementable comparison between its continuation consequences. Standard path-independence arguments convert exact switching transfers into money-valued differences; these representation steps are treated as measurement infrastructure rather than as substantive results.

The first identification result is geometric. If continuation action marginals are known on an observed family of action-generated continuation fibers, then the ambiguity in their state derivative is exactly the annihilator of a tangency space in the projectivized tangent bundle. This gives a sharp transition-matching criterion and a constructive separation proof for failure of identification.

The second result shows when recursive structure adds information. Suppose the current marginal compensation rate decomposes as an unknown flow marginal depending on state only through a known index, plus the continuation action marginal. Continuation comparisons need then be directly observed only on a seed set of inherited states. Current compensation data propagate continuation marginals from the seed set to every observed state with the same index. The resulting index saturation can strictly enlarge the transition geometry and identify off-support comparative statics that direct continuation comparisons alone leave free. If continuation fibers are already directly covered at every observed state, the index restriction adds no further continuation identification. A local optimal-choice condition then converts the sharp continuation result into a sharp comparative-static identification theorem.

1 Introduction

Fix an inherited state and compare two current actions in a binary menu. Vary a monetary transfer until choice switches. The switching amount is an observable difference in a monetary account. The paper asks what such observations reveal about an endogenous action response when current actions alter continuation consequences.

The order of analysis is deliberately revealed-preference first. Choice observations precede numerical representation, as in Samuelson (1938, 1948); Houthakker (1950). The classical integrability literature asks when observable local behavior admits a consistent numerical representation; Samuelson (1950) emphasizes both the empirical content of finite revealed-preference restrictions and the distinction between an economist's integration path and the consumer's temporal path. Hurwicz and Uzawa (1971), Afriat (1967), Diewert (1973), and Varian (1982) develop complementary demand and finite-data formulations. In the present paper, exactness of observed monetary differences is therefore background machinery, not the source of the main result.

The experiment has one monetary ruler. A current comparison (s, x, x') determines two continuation consequences through a known consequence map. If those two consequences are jointly implementable, the researcher compares them using the same monetary operation. There is no independently chosen continuation benchmark and no second monetary scale. This protocol assumption is substantive when continuation consequences differ in timing or risk: the settlement convention and account identity are held fixed across current and continuation comparisons.

The paper's first substantive result concerns continuation identification. In the deterministic smooth case, let $\Gamma(s, x)$ be the continuation state. Direct continuation comparisons at inherited state s can reveal money-valued differences along the action-generated fiber $x \mapsto \Gamma(s, x)$. If such fiber differences are known over an observed state support A , any observationally equivalent continuation valuation has zero derivative in the observed action direction on the transition image. The action direction defines a line field, hence a graph in the projectivized tangent bundle. Theorem 3.2 shows that the complete ambiguity space in the state derivative of the continuation action marginal is the annihilator of the directions whose lifted transition displacement is tangent to this graph. Transition matching is therefore necessary and sufficient, not merely sufficient. The proof uses an explicit local separation lemma rather than an appeal to unrestricted perturbations.

The second result introduces a primitive recursive restriction and asks exactly what it buys. Let $q^C(s, x)$ denote the current marginal compensation rate and let

$$q_\Psi^S(s, x) = D_x[\Psi(\Gamma(s, x))]$$

be the continuation action marginal under continuation valuation Ψ . Suppose

$$q^C(s, x) = \mu(x, \zeta(s)) + q_\Psi^S(s, x), \quad (1)$$

where ζ is a known state index and μ is otherwise unrestricted. This is stronger than a pointwise derivative exclusion. It implies cross-state restrictions over the observed support. If a state s shares its index with a directly continuation-covered state $\tilde{s}$, then

$$q_\Psi^S(s, x) = q_\Psi^S(\tilde{s}, x) + q^C(s, x) - q^C(\tilde{s}, x). \quad (2)$$

Thus current compensation data propagate continuation marginals to fibers on which no continuation comparison was directly run. This resembles the modern revealed-preference program of deriving observable cross-observation restrictions from economically specific structure rather than treating the structure as an unobservable normalization; see Echenique (2020); Echenique, Imai, and Saito (2020). Earlier nonparametric intertemporal and habit-demand analyses include Browning (1989), Crawford (2010), and Demuynck and Verriest (2013); they derive observable rationalizability restrictions for intertemporal or habit structures from demand data rather than from the monetary-comparison protocol studied here.

Equation (2) generates a sharp boundary. If continuation fibers are directly covered at every observed state, the recursive index restriction does not shrink the continuation observational-equivalence class: the within-fiber continuation differences are already fixed. Its contribution is then only to restrict the current/continuation decomposition. If direct continuation coverage is sparse, however, the same restriction can enlarge the effective family of known continuation fibers and thereby enlarge the set of transition-matched counterfactual directions. Section 6 gives a two-dimensional continuation example in which an off-support response is unidentified from direct continuation comparisons but becomes identified after index propagation.

Finally, current choice is endogenized through a local resource condition. The resulting implicit-function calculation is standard comparative statics in the sense of Samuelson (1947). The economic content lies in which state derivative of the revealed current marginal value is fixed by the combined current and continuation data. Theorem 5.1 gives the sharp set of response directions under the formally defined local recursive model class.

The compensation literature provides the closest language for the monetary representation. McKenzie (1957) develops demand theory without a utility index. Fuchs-Seliger (1989, 1990a) use income-compensation objects to obtain money-metric representations and compensated-demand results. Fuchs-Seliger (1990b) shows that income-compensation methods need not begin from complete and transitive preferences, and Fuchs-Seliger (1993) develops an equivalent compensated-demand-function approach. These results reinforce the distinction between a money-valued representation of compensated behavior and a primitive cardinal utility assumption. Quasilinear integrability, by contrast, imposes additional preference structure; see Rochet (1987); Nocke and Schutz (2017).

The identification language follows Gorno (2019): the object is an identified set of valuations and the relevant differences are observationally equivalent deviations. Local rank and tangent-space approaches to identification have classical and nonparametric precedents in Rothenberg (1971) and Chen, Chernozhukov, Lee, and Newey (2014); the present target is instead defined by the observational-equivalence class generated by compensated continuation comparisons. Dynamic choice provides related but structurally different identification problems; see Hotz and Miller (1993); Magnac and Thesmar (2002); Kalouptsi, Scott, and Souza-Rodrigues (2021); Abbring and Daljord (2020). Auer (2026) is particularly close in economic motivation: static willingness to pay may combine a contemporaneous component with continuation value, and the continuation component can be set identified. The present design instead observes selected monetary differences on continuation consequences induced by current comparisons and asks which local counterfactual response derivatives are invariant to the remaining continuation ambiguity.

2 One monetary protocol and difference data

Let the inherited state lie in a finite-dimensional smooth manifold $\mathbb{S}$, and let the current action be scalar, $x \in I \subset \mathbb{R}$. Let $\mathcal{T}_m$ denote the physical operation of adding signed amount m to the experimental monetary account, whenever feasible. The protocol fixes the account, monetary unit, and settlement convention, and satisfies

$$\mathcal{T}_m \circ \mathcal{T}_n = \mathcal{T}_{m+n}. \quad (3)$$

No preference linearity in money is imposed by (3); it specifies how transfers are recorded and combined by the experiment.

At fixed s , an oriented current comparison $e = (s, x, x')$ yields an exact switching transfer τ_e^C when the observed choice switches at a unique transfer. Regard the observed comparisons as an oriented graph. We use the standard exactness fact that edge amounts admit a node representation

$$\tau_e^C = W^R(x', s) - W^R(x, s) \quad (4)$$

if and only if every finite closed walk has zero signed transfer. The representation is unique up to an additive constant on each connected component. In a scalar locally connected action

domain, the same statement can be written as additivity of adjacent switching amounts. We use (4) only as the accounting step that defines revealed monetary differences. Where the derivative required below exists, set

$$q^C(s, x) := D_x W^R(x, s). \quad (5)$$

The current data are assumed rich enough to determine q^C on the observed state-action support used in the local analysis.

A current comparison also determines two continuation consequences. For the main results the consequence map is a deterministic transition

$$(s, x) \mapsto \Gamma(s, x) \in \mathbb{Y}. \quad (6)$$

The stochastic case can be formulated by taking the complete continuation law as the consequence, but no stochastic extension is claimed here.

For selected current comparisons, the induced pair

$$\Gamma(s, x), \quad \Gamma(s, x')$$

is jointly implemented and compared with the same monetary operation $\mathcal{T}_m$. If L_Γ denotes the physical implementation map from a current contract to its continuation consequence, the protocol requires

$$L_\Gamma(\mathcal{T}_m c) = \mathcal{T}_m(L_\Gamma c) \quad (7)$$

whenever both sides are feasible. Equation (7) is an experimental-institution assumption: it states that the continuation comparison introduces no new monetary coordinate.

The resulting continuation switching transfers τ_e^S form another oriented comparison graph on continuation consequences. Whenever its finite closed walks have zero signed transfer, choose a money-metric representation Ψ^R satisfying

$$\tau_e^S = \Psi^R(\Gamma(s, x')) - \Psi^R(\Gamma(s, x)). \quad (8)$$

Again, the root-and-path construction behind (8) is standard and will not be treated as a separate theoretical result. The substantive identification problem begins because the induced comparison graph need not determine Ψ^R away from the observed differences.

3 Continuation identification from action-generated fibers

Fix an evaluation point (s_0, x_0) and a local state support $A \subset \mathbb{S}$ through s_0 . Let A be a C^3 embedded manifold and let I be an interval around x_0 . Suppose the induced continuation comparisons determine the continuation action marginal

$$q_\Psi^S(s, x) := D_x[\Psi(\Gamma(s, x))] \quad (9)$$

for every $(s, x) \in A \times I$. A sufficient behavioral design is that, at each $s \in A$, the observed induced comparison graph is locally connected along the action-generated fiber $\Gamma(s, I)$ and the candidate valuations are C^2 .

Let $\mathfrak{V}_A$ be a nonempty local C^2 class of continuation valuations that fit the observed continuation transfers. In the local nonparametric design considered here, two members $\Psi_0, \Psi_1 \in \mathfrak{V}_A$ may differ by any C^2 function ϕ satisfying

$$D_x[\phi(\Gamma(s, x))] = 0 \quad ((s, x) \in A \times I). \quad (10)$$

Since I is connected, (10) says that ϕ is constant on each observed action-generated continuation fiber. This formally defines the observationally equivalent local perturbation class used below.

The counterfactual continuation target is

$$\Lambda_\ell(\Psi) := D_s D_x [\Psi(\Gamma(s, x))]_{(s_0, x_0)} \ell, \quad \ell \in T_{s_0} \mathbb{S}. \quad (11)$$

3.1 The action-direction lift

Assume the local transition image

$$\mathcal{M}_A := \Gamma(A \times I) \quad (12)$$

is a C^3 embedded submanifold of $\mathbb{Y}$ and $D_x \Gamma \neq 0$ near (s_0, x_0) . Suppose that whenever two observed pairs generate the same $y \in \mathcal{M}_A$, their action derivatives span the same line. Assume the resulting line field admits local C^2 nonvanishing representatives, and denote its line at y by $\mathcal{L}_y \subset T_y \mathbb{Y}$.

Let $\mathbb{P}(T\mathbb{Y})$ be the projectivized tangent bundle. Define

$$\widehat{\Gamma}(s, x) := (\Gamma(s, x), [D_x \Gamma(s, x)]), \quad (13)$$

and the graph of the observed action line field

$$\widehat{\mathcal{M}}_A := \{(y, \mathcal{L}_y) : y \in \mathcal{M}_A\}. \quad (14)$$

Assume $\widehat{\mathcal{M}}_A$ is a C^2 embedded submanifold near $\widehat{y}_0 = \widehat{\Gamma}(s_0, x_0)$. Define

$$\mathcal{C}_A := \{\ell \in T_{s_0} \mathbb{S} : D_s \widehat{\Gamma}(s_0, x_0) \ell \in T_{\widehat{y}_0} \widehat{\mathcal{M}}_A\}. \quad (15)$$

The following local separation statement is the only differential-geometric ingredient needed for sharpness.

Lemma 3.1 (Fiber-preserving separation). *All functions in this lemma are understood as local germs at $y_0 = \Gamma(s_0, x_0)$, and the fiber condition is imposed on a neighborhood of (s_0, x_0) in $A \times I$. Let $\mathcal{N}_A$ be the C^2 germs satisfying (10). Then the set of continuation-target changes*

$$\mathcal{U}_A := \{D_s D_x [\phi(\Gamma(s, x))]_{(s_0, x_0)} : \phi \in \mathcal{N}_A\} \quad (16)$$

is exactly the annihilator of $\mathcal{C}_A$:

$$\boxed{\mathcal{U}_A = \mathcal{C}_A^\perp}. \quad (17)$$

Proof. Use flow-box coordinates (r, z) on $\mathcal{M}_A$ in which the action line field is spanned by ∂_r , and extend them to ambient coordinates (r, z, n) with $\mathcal{M}_A = \{n = 0\}$. In the corresponding affine chart of $\mathbb{P}(T\mathbb{Y})$, write a representative of a nearby line as

$$\partial_r + \alpha_z \partial_z + \alpha_n \partial_n.$$

The lifted observed line field is then locally

$$\widehat{\mathcal{M}}_A = \{n = 0, \alpha_z = 0, \alpha_n = 0\}.$$

For $\phi \in \mathcal{N}_A$, define

$$F_\phi(r, z, n, \alpha) = D\phi(r, z, n)[\partial_r + \alpha_z \partial_z + \alpha_n \partial_n].$$

The fiber condition gives $F_\phi = 0$ on $\widehat{\mathcal{M}}_A$, so $dF_\phi(\widehat{y}_0)$ is conormal to $\widehat{\mathcal{M}}_A$.

Conversely, the conormal space at the origin is spanned by dn , $d\alpha_z$, and $d\alpha_n$. Given arbitrary coefficient vectors (d, b, c) , the local germ

$$\phi(r, z, n) = b \cdot z + c \cdot n + r d \cdot n$$

satisfies $\partial_r \phi(r, z, 0) = 0$ identically and yields

$$dF_\phi(\widehat{y}_0) = d \cdot dn + b \cdot d\alpha_z + c \cdot d\alpha_n.$$

Hence every conormal covector is generated by a fiber-preserving perturbation. No cutoff is needed because the statement concerns local germs.

Finally write $D_x \Gamma = \lambda v$, where v is the normalized representative used above and $\lambda(s_0, x_0) \neq 0$. Then

$$D_x[\phi \circ \Gamma] = \lambda F_\phi \circ \widehat{\Gamma}.$$

Since $F_\phi(\widehat{y}_0) = 0$ for a fiber-preserving perturbation,

$$D_s D_x[\phi \circ \Gamma]_{(s_0, x_0)} = \lambda(s_0, x_0) (D_s \widehat{\Gamma}(s_0, x_0))^* dF_\phi(\widehat{y}_0).$$

Thus $\mathcal{U}_A$ is the pullback of the conormal space to $\widehat{\mathcal{M}}_A$, up to a nonzero scalar. This pullback is the annihilator of the preimage defining $\mathcal{C}_A$ in (15). $\square$

Theorem 3.2 (Transition matching). *Under the assumptions above, the continuation target Λ_ℓ is identified over $\mathfrak{V}_A$ if and only if*

$$\boxed{\ell \in \mathcal{C}_A}. \quad (18)$$

Equivalently, the complete local ambiguity in $D_s q_\Psi^S(s_0, x_0)$ is $\mathcal{C}_A^\perp$.

Proof. For two data-compatible valuations, the target difference is generated by some $\phi \in \mathcal{N}_A$. Lemma 3.1 states that all target differences form $\mathcal{C}_A^\perp$. A direction ℓ has the same target value for every candidate valuation exactly when every covector in $\mathcal{C}_A^\perp$ annihilates ℓ , which is equivalent to $\ell \in \mathcal{C}_A$. $\square$

The projective condition is intrinsic. A coordinate expression is available under an additional rank condition rather than by definition. Suppose $\widehat{\Gamma}|_{A \times I}$ is a submersion onto $\widehat{\mathcal{M}}_A$ at (s_0, x_0) . Let $S_A = T_{s_0} A$. Then $\ell \in \mathcal{C}_A$ if and only if there exist $k \in S_A$ and $\eta \in \mathbb{R}$ such that

$$D_s \Gamma \ell = D_s \Gamma k + \eta D_x \Gamma, \quad (19)$$

$$D_s D_x \Gamma \ell - D_s D_x \Gamma k - \eta D_{xx} \Gamma \in \text{span}\{D_x \Gamma\}. \quad (20)$$

Without the submersion condition, (19)–(20) remain a sufficient parameter-space test but need not characterize the full tangent space of the image.

4 Recursive marginal structure and index propagation

We now introduce the structural restriction that links current and continuation values. Let $K \subset \mathbb{S}$ be the observed current-state support through s_0 , with K a C^2 embedded manifold, and suppose $q^C(s, x)$ is identified on $K \times I$ from the current compensated-choice data. Let

$K^d \subset K$ be a C^2 seed submanifold through s_0 on which induced continuation comparisons directly identify $q_\Psi^S(s, x)$ for all $x \in I$.

Let $\zeta : U \rightarrow \mathbb{Z}$ be a known C^2 state index on a neighborhood U of s_0 . A local recursive rationalization is a triple (q, Ψ, μ) with

$$q(s, x) = \mu(x, \zeta(s)) + D_x[\Psi(\Gamma(s, x))], \quad (s, x) \in U \times I, \quad (21)$$

$$q(s, x) = q^C(s, x), \quad (s, x) \in K \times I, \quad (22)$$

where q is C^1 , Ψ is C^2 , and μ is C^1 . The continuation valuation must also fit the directly observed continuation transfers on K^d . Denote the resulting model class by $\mathfrak{R}(K, K^d, \zeta)$ and assume it is nonempty. Equation (21) is a restriction on the current marginal residual, not a Bellman equation.

The seed design is *index-covering* for K when, locally,

$$\zeta(K) \subseteq \zeta(K^d). \quad (23)$$

Thus every observed state shares its index value with at least one directly continuation-covered state. No continuation comparison need be run at the remaining states.

Proposition 4.1 (Same-index propagation). *Suppose (23) holds. For any $s \in K$, choose $\tilde{s} \in K^d$ with $\zeta(\tilde{s}) = \zeta(s)$. Then every recursive rationalization satisfies*

$$\boxed{q_\Psi^S(s, x) = q_\Psi^S(\tilde{s}, x) + q^C(s, x) - q^C(\tilde{s}, x)}, \quad x \in I. \quad (24)$$

Consequently, the continuation action marginal is identified on all of $K \times I$. If two seed states share the same index value, the right side of (24) must be independent of which seed is used; when both sides are directly observed this is a testable cross-state restriction.

Proof. At s and $\tilde{s}$, equation (21) gives

$$q^C(s, x) - q_\Psi^S(s, x) = \mu(x, \zeta(s)),$$

$$q^C(\tilde{s}, x) - q_\Psi^S(\tilde{s}, x) = \mu(x, \zeta(\tilde{s})).$$

The right sides are equal because the two states have the same index. Rearrangement gives (24). The seed continuation marginal and both current marginals are identified by the corresponding compensated-choice data, so the right side is common to every recursive rationalization. $\square$

Proposition 4.1 converts a structural restriction into an observable propagation rule. In finite differences, the same statement is that

$$[W^R(x', s) - W^R(x, s)] - [\Psi(\Gamma(s, x')) - \Psi(\Gamma(s, x))]$$

depends on s only through $\zeta(s)$. Differentiating in x' at $x' = x$ yields (24).

Because the propagated action marginal is identified on $K \times I$, the effective transition support for Theorem 3.2 is $A = K$, even though continuation comparisons were directly run only on K^d . Write

$$\mathcal{C}_{\text{rec}} := \mathcal{C}_K, \quad \mathcal{C}_{\text{dir}} := \mathcal{C}_{K^d}, \quad (25)$$

when both regular transition geometries are defined. If the lifted direct image is an embedded submanifold of the lifted propagated image near $\hat{y}_0$, then

$$\mathcal{C}_{\text{dir}} \subseteq \mathcal{C}_{\text{rec}}. \quad (26)$$

The inclusion can be strict, as Section 6 shows.

Corollary 4.2 (No additional continuation identification under full direct coverage). *If $K^d = K$, the recursive index restriction (21) does not shrink the local continuation observational-equivalence class beyond fiber preservation on K . It may reject the model through the requirement that $q^C - q^S$ factor through ζ , but conditional on rationalizability it adds no continuation identification beyond Theorem 3.2 with $A = K$.*

Proof. With direct coverage on every state in K , two continuation valuations fitting the same continuation data have the same q^S on $K \times I$. Hence their difference already satisfies (10) on all of K . The recursive restriction can test the common residual $q^C - q^S$, but it imposes no additional condition on a continuation deviation whose action marginal is already zero on K . Off the observed support, q is an unobserved local extension and can be defined from (21). $\square$

5 Endogenous choice and sharp comparative statics

Let $\kappa(x, s)$ be the known marginal resource cost of the current action in the same monetary account. An interior local optimum satisfies

$$\kappa(x, s) = q(s, x). \quad (27)$$

At (s_0, x_0) define

$$H := D_x[\kappa - q^C](x_0, s_0), \quad (28)$$

and assume $H > 0$. The implicit-function theorem gives a unique differentiable local branch $x = g(s)$ through the point. Writing $R = D_s g(s_0)$,

$$HR = D_s q(s_0, x_0) - D_s \kappa(x_0, s_0). \quad (29)$$

This is the standard comparative-static step; identification turns on the first term on the right.

Let

$$S_0 := T_{s_0} K, \quad E := \ker D\zeta(s_0). \quad (30)$$

Current compensation data identify $D_s q$ on S_0 . The recursive structure gives, for $e \in E$,

$$D_s q(s_0, x_0)e = D_s q_\Psi^S(s_0, x_0)e = \Lambda_e(\Psi), \quad (31)$$

because the state derivative of $\mu(x, \zeta(s))$ vanishes in directions in E . By Proposition 4.1, the relevant continuation geometry is the propagated support K , so Theorem 3.2 identifies the right side of (31) exactly on $E \cap \mathcal{C}_{\text{rec}}$.

For the sharp converse, impose two local richness conditions already implicit in the model class. First, $\zeta|_K$ has constant rank and $\zeta(K)$ is an embedded submanifold near $\zeta(s_0)$. Second, apart from the observed fiber restrictions, the continuation valuation is locally nonparametric as in Section 3. These conditions define which observationally equivalent local models are being quantified over.

Theorem 5.1 (Sharp comparative-static identification). *Suppose the seed design is index-covering, the propagated transition geometry satisfies the assumptions of Theorem 3.2, and the local recursive class above is nonempty. Then the directional response $R(\ell)$ is identified if and only if*

$$\boxed{\ell \in S_0 + (E \cap \mathcal{C}_{\text{rec}})}. \quad (32)$$

Thus recursion adds off-support response directions only when the state index excludes a flow-margin change and the propagated continuation geometry transition-matches the counterfactual direction.

Proof. Sufficiency follows directly from (29). On S_0 , $D_s q$ is observed from current compensation data. On $E \cap \mathcal{C}_{\text{rec}}$, equation (31) and Theorem 3.2 identify the same derivative. Linearity then gives identification on the sum in (32).

For sharpness, compare two recursive rationalizations and let a denote the resulting change in $D_s q(s_0, x_0)$. Proposition 4.1 implies that their continuation action marginals agree on $K \times I$. Hence a continuation deviation contributes a covector in $\mathcal{C}_{\text{rec}}^\perp$, and Lemma 3.1 shows that every covector in $\mathcal{C}_{\text{rec}}^\perp$ is locally attainable.

The difference in the flow marginal is a function $\delta\mu(x, \zeta(s))$ that vanishes on $K \times I$, because both the current and propagated continuation marginals agree there. At (s_0, x_0) its state derivative therefore annihilates both S_0 and E . Conversely, the constant-rank assumption on $\zeta|_K$ implies that every covector in $(S_0 + E)^\perp$ can be generated by a local perturbation $\delta\mu(x, z) = P(x)f(z)$, with $P(x_0) = 1$ and f vanishing on the embedded set $\zeta(K)$.

The two perturbations can be chosen independently. A continuation perturbation from Lemma 3.1 has zero continuation action-marginal change on $K \times I$, so keeping μ fixed preserves the current fit on K . A flow perturbation constructed from $f|_{\zeta(K)} = 0$ keeps Ψ fixed and also preserves the current fit. Their sum therefore attains every element of

$$(S_0 + E)^\perp + \mathcal{C}_{\text{rec}}^\perp. \quad (33)$$

The directions annihilated by every covector in (33) form

$$(S_0 + E) \cap \mathcal{C}_{\text{rec}} = S_0 + (E \cap \mathcal{C}_{\text{rec}}),$$

where $S_0 \subseteq \mathcal{C}_{\text{rec}}$ because variations within the observed state support generate tangent displacements of the propagated lifted image. Equation (29) transfers the same ambiguity to the response, completing the proof. $\square$

6 A strict information gain from sparse continuation coverage

Let the inherited state be

$$s = (z, w, h) \in \mathbb{R}^3,$$

and suppose current compensation data are observed on

$$K = \{(z, w, h) : h = 0\}. \quad (34)$$

Take the known flow index to be

$$\zeta(z, w, h) = z. \quad (35)$$

Thus

$$S_0 = \text{span}\{e_z, e_w\}, \quad E = \ker D\zeta = \text{span}\{e_w, e_h\}.$$

Suppose direct continuation comparisons are run only on the seed set

$$K^d = \{(z, w, h) : w = 0, h = 0\}. \quad (36)$$

Because $\zeta(K^d) = \zeta(K)$, this sparse continuation design is index-covering. Let the continuation state be two-dimensional and let

$$\Gamma(z, w, h, x) = (x, w + h). \quad (37)$$

The action direction is $D_x\Gamma = (1, 0)$ everywhere.

Under direct continuation data alone, the observed transition image from K^d is the line

$$\mathcal{M}_{\text{dir}} = \{(x, 0) : x \in I\}.$$

At the origin,

$$D_h\Gamma = (0, 1),$$

which is transverse to this line. The action line itself does not rotate. Hence $e_h \notin \mathcal{C}_{\text{dir}}$, and direct continuation comparisons do not identify the continuation contribution to the h response. The failure is explicit: for

$$\phi_\lambda(y_1, y_2) = \lambda y_1 y_2, \quad (38)$$

we have $\phi_\lambda(x, 0) = 0$, so every direct seed-fiber transfer is preserved, while

$$D_h D_x [\phi_\lambda(\Gamma(z, w, h, x))]_0 = \lambda.$$

Now impose the recursive marginal restriction (21). Proposition 4.1 gives, for every observed $(z, w, 0)$,

$$q_\Psi^S(z, w, 0, x) = q_\Psi^S(z, 0, 0, x) + q^C(z, w, 0, x) - q^C(z, 0, 0, x). \quad (39)$$

Thus the continuation action marginal is identified on all of K , even though continuation comparisons were directly run only at $w = 0$. The propagated transition image is locally open in $\mathbb{R}^2$:

$$\mathcal{M}_{\text{rec}} = \{(x, w) : (x, w) \text{ near } (0, 0)\}.$$

Therefore $D_h\Gamma = (0, 1)$ is tangent to the propagated image and $e_h \in \mathcal{C}_{\text{rec}}$. Since also $e_h \in E$ but $e_h \notin S_0$, Theorem 5.1 identifies a genuinely off-support comparative static.

The same conclusion is visible without projective notation. Because

$$q_\Psi^S(z, w, h, x) = \Psi_1(x, w + h),$$

propagation identifies q_Ψ^S as a function of w on $h = 0$, so

$$D_h q_\Psi^S(0) = D_w q_\Psi^S(0) \quad (40)$$

is identified. The perturbation (38) is no longer admissible: on $h = 0$ it changes the continuation action marginal by λw , which cannot be offset by a flow marginal depending on state only through z . If $D_h\kappa$ is known, the response is

$$R_h = \frac{D_w q_\Psi^S(0) - D_h \kappa(0)}{H}. \quad (41)$$

This example isolates the source of the gain. The primitive state-index restriction does not identify continuation values by itself. It transfers a continuation marginal from directly covered fibers to otherwise uncovered fibers using observed current marginal differences. The enlarged transition geometry then carries that information into the off-support derivative.

7 Conclusion

Compensated switching transfers provide money-valued differences; their path-independent representation is standard measurement infrastructure. The identification problem begins when induced continuation comparisons cover only a restricted set of continuation consequences.

For a smooth deterministic transition, the action-generated continuation fibers define a line field. The projective transition-matching theorem gives the exact local continuation ambiguity and, through a separation lemma, a sharp necessity result. A primitive recursive marginal restriction can then do one of two things. With full direct continuation coverage it creates no additional continuation identification. With sparse coverage it can propagate continuation marginals across observed states that share the same state index, enlarge the effective transition geometry, and strictly identify new counterfactual response directions. The endogenous-choice step itself is standard comparative statics; the substantive content is which derivative of the revealed current marginal value is fixed by the experimental design and recursive restriction.

A Finite switching brackets

Exact switching is not required for the basic identified-set formulation. If comparison edge $e : y \rightarrow y'$ locates the switching transfer only in

$$[\underline{\tau}_e, \bar{\tau}_e],$$

a compatible monetary representation satisfies

$$\underline{\tau}_e \leq \Psi(y') - \Psi(y) \leq \bar{\tau}_e. \quad (42)$$

The identified set is then defined by difference inequalities rather than equalities. The sharp transition-matching theorem above is stated for exact local fiber differences; extending its local separation statement to binding and slack inequality patterns requires keeping track of the baseline valuation and is not pursued here.

B Recursive payoff representation

Whenever current and continuation money metrics are jointly defined on a local domain, one may define the residual money-valued flow object

$$u^R(x, s) := W^R(x, s) - \Psi^R(\Gamma(s, x)). \quad (43)$$

Then

$$W^R(x, s) = u^R(x, s) + \Psi^R(\Gamma(s, x)) \quad (44)$$

is an accounting identity. A stationary Bellman equation would require additional recursive closure, a discounting convention, and a fixed-point interpretation; none is needed for the identification results in this paper.

References

- Abbring, J. H. and O. Daljord (2020). “Identifying the Discount Factor in Dynamic Discrete Choice Models.” *Quantitative Economics* 11, 471–501.
- Afriat, S. N. (1967). “The Construction of Utility Functions from Expenditure Data.” *International Economic Review* 8, 67–77.
- Auer, Josephine (2026). “When Does Static Willingness to Pay Mislead? A Framework for Dynamic Hedonic Valuation.” Working paper, MIT, Version 6, August 21, 2026.
- Diewert, W. E. (1973). “Afriat and Revealed Preference Theory.” *Review of Economic Studies* 40, 419–425.
- Echenique, F. (2020). “New Developments in Revealed Preference Theory: Decisions Under Risk, Uncertainty, and Intertemporal Choice.” *Annual Review of Economics* 12, 299–316.
- Echenique, F., T. Imai, and K. Saito (2020). “Testable Implications of Models of Intertemporal Choice: Exponential Discounting and Its Generalizations.” *American Economic Journal: Microeconomics* 12(4), 114–143.
- Fuchs-Seliger, S. (1989). “Money-Metric Utility Functions in the Theory of Revealed Preference.” *Mathematical Social Sciences* 18, 199–210.
- Fuchs-Seliger, S. (1990a). “An Axiomatic Approach to Compensated Demand.” *Journal of Economic Theory* 52, 111–122.
- Fuchs-Seliger, S. (1990b). “Compensated and Direct Demand Without Transitive and Complete Preferences.” *Annals of Operations Research* 23, 299–310.
- Fuchs-Seliger, S. (1993). “A Reformulation of the Theory of Demand by Compensated Demand Functions.” *Journal of Economics* 57, 279–293.
- Browning, M. (1989). “A Nonparametric Test of the Life-Cycle Rational Expectations Hypothesis.” *International Economic Review* 30, 979–992.
- Chen, X., V. Chernozhukov, S. Lee, and W. K. Newey (2014). “Local Identification of Nonparametric and Semiparametric Models.” *Econometrica* 82, 785–809.
- Crawford, I. (2010). “Habits Revealed.” *Review of Economic Studies* 77, 1382–1402.
- Demuyndck, T. and E. Verriest (2013). “I’ll Never Forget My First Cigarette: A Revealed Preference Analysis of the ‘Habits as Durables’ Model.” *International Economic Review* 54, 717–738.
- Gorno, L. (2019). “Revealed Preference and Identification.” *Journal of Economic Theory* 183, 698–739.
- Hotz, V. J. and R. A. Miller (1993). “Conditional Choice Probabilities and the Estimation of Dynamic Models.” *Review of Economic Studies* 60, 497–529.
- Houthakker, H. S. (1950). “Revealed Preference and the Utility Function.” *Economica* 17, 159–174.

- Hurwicz, L. and H. Uzawa (1971). “On the Integrability of Demand Functions.” In J. S. Chipman, L. Hurwicz, M. K. Richter, and H. F. Sonnenschein (eds.), *Preferences, Utility, and Demand*, Harcourt Brace Jovanovich, New York, 114–148.
- Kalouptsi, M., P. T. Scott, and E. Souza-Rodrigues (2021). “Identification of Counterfactuals in Dynamic Discrete Choice Models.” *Quantitative Economics* 12, 351–403.
- Magnac, T. and D. Thesmar (2002). “Identifying Dynamic Discrete Decision Processes.” *Econometrica* 70, 801–816.
- McKenzie, L. W. (1957). “Demand Theory Without a Utility Index.” *Review of Economic Studies* 24, 185–189.
- Nocke, V. and N. Schutz (2017). “Quasi-linear Integrability.” *Journal of Economic Theory* 169, 603–628.
- Rochet, J.-C. (1987). “A Necessary and Sufficient Condition for Rationalizability in a Quasi-linear Context.” *Journal of Mathematical Economics* 16, 191–200.
- Rothenberg, T. J. (1971). “Identification in Parametric Models.” *Econometrica* 39, 577–591.
- Samuelson, P. A. (1938). “A Note on the Pure Theory of Consumer’s Behaviour.” *Economica* 5, 61–71.
- Samuelson, P. A. (1947). *Foundations of Economic Analysis*. Harvard University Press, Cambridge, MA.
- Samuelson, P. A. (1948). “Consumption Theory in Terms of Revealed Preference.” *Economica* 15, 243–253.
- Samuelson, P. A. (1950). “The Problem of Integrability in Utility Theory.” *Economica* 17, 355–385.
- Varian, H. R. (1982). “The Nonparametric Approach to Demand Analysis.” *Econometrica* 50, 945–973.